

Anurag Satpathy

Postdoctoral Fellow, CReWMaN Lab, Department of Computer Science

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Research Focus

My research focuses on **Intelligent Cyber-Physical Systems (CPS)** for distributed resource coordination in dynamic, resource-constrained edge/IoT environments. I model strategic interactions among self-interested agents accessing shared resources under real-time constraints, uncertain network conditions, heterogeneous user behavior, and application-specific operational challenges. Using game-theoretic and learning-based methods, I design new resource-allocation mechanisms, establish equilibrium conditions, and validate their performance across mobility-driven, safety-critical, and human-in-the-loop CPS applications. My work contributes to the design of mechanisms and equilibria for distributed CPS while addressing practical problems in intelligent mobility, EV charging, UAV-enabled healthcare logistics, and public safety.

Research Interests

- Mechanism design for distributed resource coordination in CPS.
- Equilibrium analysis of strategic agents sharing constrained resources.
- Game-theoretic and learning-based resource allocation under uncertainty.
- Stable matching, scheduling, and reassignment in CPS.
- Behavior-aware resource coordination for human-in-the-loop CPS.

Education

- **Ph.D. in Computer Science and Engineering** Aug 2017 – Feb 2023
National Institute of Technology (NIT) Rourkela, India
Institute of National Importance; ranked 13th in Engineering in NIRF 2025, Government of India
Dissertation: *Efficient Re-Embedding Strategies for Virtual Data Centers over Multi-Domain Substrate Networks*
- **M.Tech. in Computer Science and Engineering** Jul 2015 – Jul 2017
Birla Institute of Technology (BIT) Mesra, India, (*Deemed-to-be University (UGC), India*)
- **B.Tech. in Information Technology** Aug 2010 – Jun 2014
International Institute of Information Technology (IIIT) Bhubaneswar, India, (*State Public University (Govt. of Odisha)*)

Professional Appointments

- **Missouri University of Science and Technology, USA** MO, USA
Postdoctoral Fellow, CReWMaN Lab, Department of Computer Science Aug 2023–Present
Mentor: Prof. Sajal K. Das
- **National Institute of Technology, Rourkela** Odisha, India
Graduate Research Assistant Fall 2017 - Spring 2023
- **National Institute of Technology, Rourkela** Odisha, India
Visiting Researcher May 2016–April 2017

External Funding & Proposal Activities

Funded Grants and Awards

- **NSF National I-Corps Program** USA
Technical Lead Accepted for Fall 2026
Project: I-Haul: Intelligent Aerial Logistics for Organ Transport
Award: \$50,000
 - Leading national customer discovery, market validation, and technology-readiness planning for autonomous organ-transport logistics.
 - Expanding stakeholder discovery to 100+ interviews across OPOs, transplant centers, FAA/UAS regulatory bodies, UAV logistics providers, and clinical users.
 - Preparing the project for SBIR/STTR translation pathways.

- **NSF I-Corps BOOST Program** USA
○ *Technical Lead* Oct 2025 – Dec 2025
Project: I-Haul: Intelligent Aerial Logistics for Organ Transport
Award: \$3,000
- Led a two-Ph.D.-student team on commercialization strategy for autonomous organ-transport drones.
 - Completed 75+ stakeholder interviews with OPOs, transplant centers, surgeons, courier companies, UAV manufacturers, researchers, and FAA stakeholders.
 - Filed provisional patent Application No. 63/860,303 on Sept. 15, 2025.

- **NSF I-Corps Local + Regional Programs** USA
○ *Technical Lead* Sep 2024 – Nov 2024
Project: I-Haul: Intelligent Aerial Logistics for Organ Transport
Award: \$1,000
- Led 25+ customer discovery interviews with clinicians, transplant teams, researchers, and UAV logistics partners.
 - Identified clinical workflow constraints, cold-chain requirements, regulatory risks, and UAV logistics barriers for emergency organ delivery.

Submitted Proposal

- **NSF ENG/ECCS/EPCN** USA
○ *Co-PI* Submitted Feb 2026
Project: SPICE: Subscription-based Platform for Intelligent Charging of Electric Vehicles
Requested Amount: \$675,597
- Designed behavior-aware EV charging assignment policies for fair and predictable charging access under congestion.
 - Modeled uncertainty, strategic user behavior, and misuse in shared charging infrastructure.
 - Studied how charging delays propagate into mobility and logistics systems.

Proposal in Development

- **NSF CISE/CCF – Algorithmic Foundations** USA
○ *Co-PI* Planned submission: Aug 2026
Project: Strategic Light-weight Reassignment of Shared Resources in Large-Scale CPS
Estimated Requested Amount: \$600,000
- Designing reassignment mechanisms for strategic agents sharing constrained resources under noncompliance, uncertainty, and dynamic preferences.
 - Establishing stability and equilibrium conditions for multi-quota resource exchange.
 - Parallel resolution of resource exchanges for re-assignments in large systems.

Selected Publications (Research Leadership)

Journal Articles

- [1] Arindam Khanda*, **Anurag Satpathy***, and Sajal K. Das. SMART-CHARGE: Stable matching algorithm for electric vehicle charging in subscription-based models. *Pervasive and Mobile Computing*, page 102197, 2026. ➡ DOI (***Equal contribution**).
- [2] **Anurag Satpathy**, Manmath Narayan Sahoo, Chittaranjan Swain, Paolo Bellavista, Khan Muhammad, and Sambit Bakshi. Virtual Network Embedding: Literature Assessment, Recent Advancements, Opportunities, and Challenges. *IEEE Communications Surveys & Tutorials*, 27(6):3861–3914, 2025. Q1; IF – 46.7; ➡ DOI.
- [3] **Anurag Satpathy**, Manmath Narayan Sahoo, Chittaranjan Swain, Muhammad Bilal, Sambit Bakshi, and Houbing Song. GAMap: A Genetic Algorithm-Based Effective Virtual Data Center Re-Embedding Strategy. *IEEE Transactions on Green Communications and Networking*, 8(2):791–801, 2024. Q1; ➡ DOI.
- [4] **Anurag Satpathy**, Manmath Narayan Sahoo, Lucky Behera, and Chittaranjan Swain. ReMatch: An Efficient Virtual Data Center Re-Matching Strategy Based on Matching Theory. *IEEE Transactions on Services Computing*, 16(2):1373–1386, 2023. Q1; ➡ DOI.
- [5] **Anurag Satpathy**, Manmath Narayan Sahoo, Arun Kumar Sangaiah, Chittaranjan Swain, and Sambit Bakshi. CoMap: An efficient virtual network re-mapping strategy based on coalitional matching theory. *Computer Networks, Elsevier*, 216:109248, 2022. Q1; ➡ DOI.
- [6] **Anurag Satpathy**, Manmath Narayan Sahoo, Ashutosh Mishra, Banshidhar Majhi, Joel J. P. C. Rodrigues, and Sambit Bakshi. A Service Sustainable Live Migration Strategy for Multiple Virtual Machines in Cloud Data Centers. *Big Data Research, Elsevier*, 25:100213, 2021. Q1; ➡ DOI.
- [7] **Anurag Satpathy**, Sourav Kanti Addya, Ashok Kumar Turuk, Banshidhar Majhi, and Gadadhar Sahoo. Crow search based virtual machine placement strategy in cloud data centers with live migration. *Computers & Electrical Engineering*, 69:334–350, 2018. Q1; ➡ DOI.

Conference Publications

- [1] **Anurag Satpathy**, Arindam Khanda, Chittaranjan Swain, and Sajal K. Das. ReACT-TTC: Capacity-Aware Top Trading Cycles for Post-Choice Reassignment in Shared CPS. In *Proceedings of the 17th ACM/IEEE International Conference on Cyber-Physical Systems (ICCPs)*, Saint-Malo, France, May 2026. Accepted for publication.
- [2] Arindam Khanda*, **Anurag Satpathy***, Amit Jha, and Sajal K. Das. CARGO: A Co-Optimization Framework for EV Charging and Routing in Goods Delivery Logistics. In *2025 IEEE 50th Conference on Local Computer Networks (LCN)*, 2025. ➡ DOI (***Equal contribution.**).
- [3] **Anurag Satpathy**, Manmath Narayan Sahoo, Lipsa Chottray, Banshidhar Majhi, Ashutosh Mishra, and Sambit Bakshi. VRMap: A Cost and Time Aware Remapping of Virtual Data Centres Over a Geo-distributed Infrastructure. In *2020 International Conference on COMMunication Systems & NETworkS (COMSNETS)*, pages 427–434, 2020. ➡ DOI.
- [4] **Anurag Satpathy**, Manmath Narayan Sahoo, Lucky Behera, Chittaranjan Swain, and Ashutosh Mishra. VMatch: A Matching Theory Based VDC Reconfiguration Strategy. In *2020 IEEE 13th International Conference on Cloud Computing (CLOUD)*, pages 133–140, 2020. ➡ DOI (**YESC Student Paper Participation Award**).

Book Chapters

- [1] **Anurag Satpathy** and Chittaranjan Swain. Strategic Decision-Making Architectures for 6G-Enabled Cyber-Physical Systems. In *Secure and Responsible AI in 6G Technology-Enabled IoT Systems*. CRC Press, Taylor & Francis Group, 2026.
- [2] **Anurag Satpathy**. Evolution of Cloud-Fog-IoT Interconnection Networks. In *Cloud Network Management*, pages 3–14. CRC Press, Taylor & Francis Group, 2020. ➡ DOI.

Additional Peer-Reviewed Publications (Selective)

Journal Articles

- [1] Chittaranjan Swain, Manmath Narayan Sahoo, **Anurag Satpathy**, and Sambit Bakshi. Fair-Match: Fairness-aware Matching Framework for Task Offloading in IoT-Fog Systems. *IEEE Transactions on Consumer Electronics*, 2026. ➡ DOI.
- [2] Priyanka Ashok Birajdar, Kush Anchalia, **Anurag Satpathy**, and Sourav Kanti Addya. RUSH: Rule-Based Scheduling for Low-Latency Serverless Computing. *IEEE Networking Letters*, pages 1–1, 2025. ➡ DOI.
- [3] Chittaranjan Swain, Manmath Narayan Sahoo, **Anurag Satpathy**, Sambit Bakshi, and Soumya K. Ghosh. M-DAFTO: Multi-Stage Deferred Acceptance Based Fair Task Offloading in IoT-Fog Systems. *IEEE Transactions on Services Computing*, 17(6):3928–3941, 2024. Q1; ➡ DOI.
- [4] Chittaranjan Swain, Manmath Narayan Sahoo, **Anurag Satpathy**, Khan Muhammad, Sambit Bakshi, and Joel J. P. C. Rodrigues. A-DAFTO: Artificial Cap Deferred Acceptance-Based Fair Task Offloading in Complex IoT-Fog Networks. *IEEE Transactions on Consumer Electronics*, 69(4):914–926, 2023. IF-10.9; ➡ DOI.
- [5] Sourav Kanti Addya, **Anurag Satpathy**, Bishakh Chandra Ghosh, Sandip Chakraborty, Soumya K. Ghosh, and Sajal K. Das. Geo-Distributed Multi-Tier Workload Migration Over Multi-Timescale Electricity Markets. *IEEE Transactions on Services Computing*, 16(5):3385–3396, 2023. Q1; ➡ DOI.
- [6] Sourav Kanti Addya, **Anurag Satpathy**, Bishakh Chandra Ghosh, Sandip Chakraborty, Soumya K. Ghosh, and Sajal K. Das. CoMCLoud: Virtual Machine Coalition for Multi-Tier Applications Over Multi-Cloud Environments. *IEEE Transactions on Cloud Computing*, 11(1):956–970, 2023. Q1; ➡ DOI.

Conference Publications

- [1] Sowjanya Tammali, Arindam Khanda, **Anurag Satpathy**, S. M. Shovan, and Sajal K. Das. RESCUE: Routing under Evolving Stochastic Congestion and Uncertain Spread in Wildfire Emergencies. In *Proceedings of the 27th International Conference on Distributed Computing and Networking (ICDCN)*, page 168–172, 2026. ➡ DOI.
- [2] Arindam Khanda, **Anurag Satpathy**, Anusha Vangala, and Sajal K. Das. SMEVCA: Stable Matching-based EV Charging Assignment in Subscription-Based Models. In *26th International Conference on Distributed Computing and Networking (ICDCN)*, 2025. ➡ DOI (**Best Paper Nomination**).
- [3] Mahmuda Akter, Debjoyti Sengupta, **Anurag Satpathy**, and Sajal K. Das. MIME: Mobility-Induced Dynamic Matching for Partial Offloading in Vehicular Edge Computing. In *2024 IEEE 49th Conference on Local Computer Networks (LCN)*, pages 1–9, 2024. ➡ DOI.
- [4] Debjoyti Sengupta, **Anurag Satpathy**, and Sajal K. Das. POSCA: Path Optimization for Solar Cover Amelioration in Urban Air Mobility. In *2024 IEEE International Conference on Smart Computing (SMARTCOMP)*, pages 6–13, 2024. ➡ DOI.
- [5] Priyanka Soni, Omkar Chand Deshlahre, **Anurag Satpathy**, and Sourav Kanti Addya. LBA: Matching Theory Based Latency-Sensitive Binary Offloading in IoT-Fog Networks. In *2024 16th International Conference on COMMunication Systems & NETworkS (COMSNETS)*, pages 165–170, 2024. ➡ DOI (**Best Paper Award**).

- [6] Aastik Verma, **Anurag Satpathy**, Sajal. K. Das, and Sourav Kanti Addya. LEASE: Leveraging Energy-Awareness in Serverless Edge for Latency-Sensitive IoT Services. In *2024 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*, pages 302–307, 2024. ➡ DOI.
- [7] Chittaranjan Swain, Manmath Narayan Sahoo, and **Anurag Satpathy**. LETO: An Efficient Load Balanced Strategy for Task Offloading in IoT-Fog Systems. In *2021 IEEE International Conference on Web Services (ICWS)*, pages 459–464, 2021. ➡ DOI.
- [8] Arpana Chakma, Shubham Kumar, Pradeep Kumar Mahato, **Anurag Satpathy**, and Sourav Kanti Addya. Matchcloud: Service matching for multi cloud marketplace. In *2021 International Conference on COMMunication Systems & NETWORKS (COMSNETS)*, pages 127–130, 2021. ➡ DOI (**Best Paper Award**).
- [9] Bishakh Chandra Ghosh, Sourav Kanti Addya, **Anurag Satpathy**, Soumya K. Ghosh, and Sandip Chakraborty. Towards a democratic federation for infrastructure service provisioning. In *2019 IEEE International Conference on Services Computing (SCC)*, pages 162–166, 2019. ➡ DOI.

Research Experience

- **Missouri University of Science and Technology** **Missouri, USA**
Postdoctoral Fellow *Aug 2023–Present*
 - Developed a new independent research direction on mechanism design, equilibrium analysis, and scalable reassignment for strategic resource coordination across CPS applications.
 - Contributed as Co-PI to the NSF ENG/ECCS/EPCN proposal *SPICE: Subscription-based Platform for Intelligent Charging of Electric Vehicles*. Led the development of two research thrusts: mechanism design for an end-to-end subscription model for EV charging, and modeling the impact of uncertain human behavior on charging systems and how they propagate through mobility and logistics systems.
 - Independently led the development of a capacity-aware exchange mechanism that fundamentally extends the Top Trading Cycles algorithm for post-choice reassignment among strategic agents. This work opened a new research direction on mechanism design and equilibrium analysis for shared-resource CPS.
 - Led research on the NSF *SOTERIA* project, developing coalition-formation and regret-learning methods for UAV-assisted emergency communication under uncertain demand and network conditions.
 - Served as Technical Lead for NSF I-Corps Local, Regional, and BOOST programs on *I-Haul*, and will serve as Technical Lead for the accepted NSF National I-Corps program; led stakeholder discovery, commercialization planning, provisional patent activity, and SBIR/STTR pathway development.
 - Directed student-led implementations on mobility-aware vehicular edge offloading, stochastic evacuation routing, and UAV-enabled organ transport; guided students through problem formulation, algorithm design, evaluation, and publication writing.
- **National Institute of Technology (NIT), Rourkela** **Odisha, India**
Graduate Research Assistant *2017–2023*
 - Independently pioneered the integration of game-theoretic approaches into virtual network reassignment for strategic resource reallocation in multi-domain substrate networks under dynamic workload conditions.
 - Introduced fairness-aware multi-quota task offloading mechanisms for shared resource allocation in IoT-fog systems under resource and latency constraints.
 - Co-mentored junior graduate and undergraduate students on implementation, simulation, and experimental evaluation.
 - Published independent and collaborative research outcomes in flagship conferences, such as IEEE Services (CLOUD, ICWS, SCC) and journals (IEEE TSC, TCE, TGCN, IoTJ).
 - Played a critical role in a collaborative research effort with Dr. Sourav Kanti Addya during his postdoctoral research under Dr. Sandip Chakraborty and Dr. Soumya K. Ghosh at IIT Kharagpur, resulting in publications in IEEE TSC and IEEE TCC.
- **National Institute of Technology, Rourkela** **Odisha, India**
Visiting Researcher *May 2016–Apr 2017*
 - Led the development of resource-efficient VM placement and migration heuristics for geo-distributed cloud data centers, resulting in a publication in *Computers & Electrical Engineering, Elsevier*.

Teaching and Mentoring Experience

Research Mentoring and Teaching Contributions (Postdoctoral)

- Mentored Ph.D. students on strategic decision-making in cyber-physical systems, including stochastic wildfire evacuation routing under selfish user behavior, UAV-assisted medical logistics and organ transportation, and mobility-aware vehicular edge offloading in highly dynamic environments.
- Delivered guest lectures at the Missouri University of Science and Technology on matching theory and strategic decision-making in CPS, Top-Trading Cycles and game-theoretic coordination for EV charging systems, and graph-theoretic approaches for distributed CPS coordination.

Teaching Experience (Ph.D.)

2018–2023: Graduate Teaching Assistant, National Institute of Technology Rourkela, Odisha, India

Courses Assisted:

- **Advanced Data Structures (CS 6173)** — Graduate
- **Operating Systems (CS 3071)** — Undergraduate

Instructional Contributions:

- Led laboratory sessions, recitations, and office hours for undergraduate and graduate students.
- Designed and updated programming assignments and laboratory modules for Operating Systems and Advanced Data Structures.
- Assisted in course redesign and integrated systems-oriented research examples into instruction.
- Evaluated programming assignments, quizzes, examinations, and project implementations.

Courses Ready to Teach: Operating Systems, Distributed Systems, Computer Networks, Cloud and Edge Computing, Algorithms, Graph Theory, Internet of Things (IoT), Cyber-Physical Systems, and Computer Systems.

Potential New Courses: Strategic interaction in CPS Applications (Application-driven course); Resource orchestration in modern CPS: Modeling, Solutions, and Evaluation (Theory-driven course).

Invited Talks

- **SRM University**, Andhra Pradesh, India Jan 2026
Invited Speaker, Online Faculty Development Program (FDP)
Theme: *AI-Driven Distributed Ecosystems: Bridging Cloud, Fog, and Edge for Next-Generation Innovation*
- **Siddaganga Institute of Technology**, Tumkur, India Jan 2025
Invited Speaker, AICTE-ATAL Academy Faculty Development Program
Theme: *Empowering Smart Cities with AI-Driven Edge and Fog Computing Architecture*

Professional Activities

Editorial, Panels, and Leadership

- Publication Co-Chair @ ICDCN 2027
- Panelist, NSF I-Corps Conference Panel on Customer Discovery Conversations, Feb 2026
- Panelist, Entrepreneur Week 2026, Missouri S&T, on the pathway to National I-Corps, April 2026
- Organizing Co-Chair, ASCENT Workshop @ ICDCN 2026
- Co-Guest Editor, *Ad Hoc Networks* (Elsevier), Special Issue on *Edge and Serverless Computing*, 2025
- Posters Co-Chair, IEEE ICDDDS 2025

Program Committee Memberships

- **2026:** IEEE/ACM International Symposium on Quality of Service (IWQoS); IEEE MASS, Systems and Applications Track; IEEE SmartGridComm, Symposium on Communication and Networking; VTC-Fall Workshop on Federated Learning for Autonomous and Connected Vehicles; ICCCN Workshop on Distributed Communication, Sensing, and Computing in Space-Air-Ground Networks for 6G; IFIP Networking CREATIVE Workshop; ICFAT; IEEE GlobeCom SAC MLCN; VTC-Spring Workshop on LPWAN-Based Terrestrial-Non-Terrestrial Integration for Vehicular Networks; GameSec; IEEE COMSNETS Posters and Graduate Forum; IEEE LCN; IEEE CONNECT; UBICOMM; IEEE PerCom SPT-IoT and EDGE-Well; IEEE COMSNETS MINDS; DCOSS-IoT Wi-DroIT.
- **2025:** IEEE LCN; IEEE CONNECT; UBICOMM; IEEE ICDCN Posters & Demos PC; IEEE PerCom SPT-IoT and EDGE-Well; IEEE COMSNETS MINDS; HiPC Student Research Symposium; DCOSS-IoT Wi-DroIT; IEEE ICDDDS.
- **2024:** IEEE COMSNETS MINDS; IEEE ICDDDS.
- **2023–2022:** IEEE ICDCN Posters & Demos PC (2023); IEEE/ACM CCGRID STEERS Workshop (2022); IEEE ICRAIE (2022).

Journal, Conference, and Proposal Reviewing

- **Proposal Reviewing:** External Reviewer, NSERC Discovery Grant (Canada).
- **Journal Reviewing:** IEEE Transactions on TUSC, T-ITS, TCE, TMC, TNET, TSC, TC, and IoT-J; ACM ToSN; Elsevier FGCS, Computer Networks, Computer Communications, JNCA, Ad Hoc Networks, and SIMPAT; Springer Cluster Computing, Grid Computing, Supercomputing, and Cloud Computing; Wiley Concurrency and Computation Practices.
- **Conference Reviewing:** IEEE ANTS 2024; IEEE INDICON 2025; IEEE BHTC 2025–2026; SCI 2025.

Awards

- 2026 NSF Grant for Presenting a paper at ICCPS, St. Malo, France
- 2025 NSF I-Corps BOOST Travel Grant for customer discovery.
- 2025 Best Paper Nomination, ACM ICDCN.
- 2024 Best Paper award, MINDS Workshop, IEEE COMSNETS.
- 2021 Best Poster Award, IEEE COMSNETS.
- 2020 YESC Student Paper Participation Award, IEEE Cloud.

References

- **Dr. Sajal K. Das**
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